



Conditional Wasserstein Autoencoders and Deep Semantic Representations for Time Series Modeling.

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Summary

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2 Litterature Review

3 Methodology

- Encoders
- Sentence Embedding and Semantic Axe
- Projection and Evaluation
- WAE-LSTM

4 Key Findings

- Forward Guidance score
- Results for multi-horizon

5 Perspectives and Discussions

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Background:

- Central banks increasingly use language as a policy tool.
- FG aims to shape market expectations by signaling future monetary policy.
- Yet, quantifying its semantic content and linking it to macro-financial variables remains challenging.

Central Question:

- How does the communication of European Central Bank, specifically through Forward Guidance (FG), impact systemic financial stress?

Goal of this Study:

- Build a continuous FG indicator based on the semantic content of ECB speeches.
- Compare different NLP models (ModernBERT, Mamba, etc.) to extract policy stance.
- Assess whether FG helps explain variations in systemic stress using conditional VAE-LSTM models.

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Forward Guidance and Systemic Stress

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- **Forward Guidance** emerged as a key monetary policy tool when rates hit the zero lower bound (Woodford, 2012).
- **Transmission channels:**
 - Interest rate expectations (Coenen et al., 2021)
 - Financial conditions (Swanson, 2021)
 - Credit markets (Bernanke and Gertler, 1995)
 - Signaling and credibility (Nakamura and Steinsson, 2018)
- **FG and systemic stress:**
 - Stabilizing with credible signals (De Haan, 2025)
 - Destabilizing under uncertainty (Linta, 2024)

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Data

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Dataset	Period	Frequency	Description
ECB Text Corpus	Jan 2008 – Dec 2024	Monthly	Press releases, speeches and minutes scraped from ECB website; preprocessed into token sequences.
CISS Index	Jan 2008 – Dec 2024	Monthly	Composite Indicator of Systemic Stress (CISS) from Brunnermeier et al., capturing financial market stress.

Table: Summary of data sources used in the study

ModernBERT: A Next-Generation Encoder

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What makes ModernBERT stand out?

- Modernized architecture: GeGLU activations, rotary positional embeddings (RoPE), and pre-layer normalization for training stability and better generalization (Warner et al., 2024).
- Long-context native support: Trained on sequences up to 8192 tokens, outperforming previous BERT variants on long-document retrieval and classification tasks.
- High efficiency: Alternating global/local attention and unpadding techniques make inference fast and memory-efficient. Using flash attention v2 and v3 :

$$\text{Attention} = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V, \quad \text{output}_i = \frac{o_i}{s_i} = \frac{\sum_{\text{blocks } b} \sum_{j \in b} \exp \left(\frac{q_i^T k_j}{\sqrt{d}} - m_i \right) v_j}{\sum_{\text{blocks } b} \sum_{j \in b} \exp \left(\frac{q_i^T k_j}{\sqrt{d}} - m_i \right)}$$

- Broad pretraining corpus: 2 trillion tokens including web, code, and scientific data enabling robust performance across domains.
- State-of-the-art results: Outperforms BERT, RoBERTa, and DeBERTaV3 on GLUE, BEIR, CodeSearchNet, and MLDR (long-context retrieval).

Mamba architecture

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Why Mamba for Forward Guidance?

- Mamba is a recent state-space model (Gu and Dao, 2024) that enables long-range dependency modeling in text.
- Offers faster inference and better memory scaling than standard transformers.
- Well-suited to capturing slow-moving semantic structures in policy documents.
- Mamba2 (Dao and Gu, 2024) extends Mamba with adaptive span mechanisms and learnable state transitions, improving stability and performance on very long sequences.
- LLaMba integrates cross-attention and hierarchical memory for conditional generation, making it ideal for combining Forward Guidance signals with textual policy analysis (Wang et al., 2025).

Selective State Space Models and Mamba: A New Attention

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The general SSM formulation is **schema** :

$$\frac{dy(t)}{dt} = \mathbf{A}y(t) + \mathbf{B}u(t) \Rightarrow \mathbf{y}_t = \mathbf{A}_t\mathbf{y}_{t-1} + \mathbf{B}_t\mathbf{x}_t$$

Discretization over time (Euler or matrix exponential):

$$\mathbf{y}_t = e^{\mathbf{A}\Delta t}\mathbf{y}_{t-1} + \mathbf{B}_t\mathbf{x}_t \quad \text{where } \mathbf{x}_t = \mathbf{u}(t)$$

Mamba: Input-dependent State Space Model

$$\mathbf{A}_t = \text{diag}(\alpha_t) = \exp(-\Delta t \cdot \mathbf{D}_t), \quad \text{learned or gated}$$

$$\mathbf{B}_t = \text{data-dependent gating: } \mathbf{B}_t = \mathbf{C}(\mathbf{x}_t)$$

$$\mathbf{y}_t = \mathbf{A}_t\mathbf{y}_{t-1} + \mathbf{B}_t\mathbf{x}_t$$

Output:

$$\mathbf{h}_t = \mathbf{W}_{\text{out}} \cdot \mathbf{y}_t$$

Mamba vs Transformers

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Property	Transformers	Mamba (SSM)
Core Mechanism	Attention (global)	State-space recurrence
Time Complexity	$\mathcal{O}(L^2)$	$\mathcal{O}(L)$
Memory Usage	High	Low
Long-Context Handling	Limited (fixed window)	Excellent (continuous state)
Sequential Processing	Parallelizable	Recurrent (streamable)
Data Dependence	Static weights	Input-conditioned dynamics
Suitability for Policy Texts	Good	Excellent (long-range structure)

Sentence Embedding and Semantic Axe

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Given a speech or sentence x in a corpus document Θ . Let a sentence be tokenized into L tokens: $x = (t_1, t_2, \dots, t_L)$. Each encoder (one-shot) associates each token t_i with a vector $h_i \in \mathbb{R}^d$ through its internal layers. We have: $H = [h_1, h_2, \dots, h_L]^T \in \mathbb{R}^{L \times d}$ where each row corresponds to the embedding of a token. For reduce in $1 \times d$ vector :

$$\mathcal{E}_x = \text{MeanPooling}(\text{Encoder}(x)) \in \mathbb{R}^d, \quad \forall x \in \Theta$$

$$\text{where MeanPooling is } \mathcal{E}_x = \frac{1}{L} \sum_{i=1}^L h_i \in \mathbb{R}^d \text{ (arithmetic mean of the vectors } h_i).$$

We then define, using three LLMs (Gemini, Mistral 7B, and Llama 3), the sets of anchor sentences :

- 1 $\mathcal{P}_{\text{exp}} = \{x_i^+\}_{i=1}^N$: expansive statements *"The Governing Council stands ready to lower interest rates further if necessary"*.
- 2 $\mathcal{P}_{\text{exp}} = \{x_i^0\}_{i=1}^N$: neutral statements *"The ECB keep interest rates unchanged"*.
- 3 $\mathcal{P}_{\text{res}} = \{x_j^-\}_{j=1}^M$: restrictive statements *"ECB increase interest rates"*.

Then, compute the semantic direction :

$$\mathcal{V} = \frac{1}{N} \sum_{i=1}^N \mathcal{E}_{x_i^+} - \frac{1}{M} \sum_{j=1}^M \mathcal{E}_{x_j^-} \in \mathbb{R}^d, \quad \hat{\mathcal{V}} = \frac{\mathcal{V}}{\|\mathcal{V}\|} \quad (\text{unit norm})$$

Sentence Embedding and Semantic Axe

14b/44

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Projection

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Each new sentence is scored via cosine projection:

$$s_x = \cos(\mathcal{E}_x, \hat{\mathcal{V}}) = \frac{\mathcal{E}_x^\top \hat{\mathcal{V}}}{\|\mathcal{E}_x\| \cdot 1}$$

This yields a scalar score reflecting how "expansive" vs. "restrictive" or "neutral" the sentence is.

We calibrate the raw scores s_x using anchor sets to compute a centered score:

$$\mu_{\text{anchor}}, \sigma_{\text{anchor}} = \text{mean}, \text{std}(\{s_{x_i^+}\} \cup \{s_{x_j^-}\}).$$

$$z_x = \frac{s_x - \mu_{\text{anchor}}}{\sigma_{\text{anchor}}}$$

Result: a normalized indicator $z_x \in \mathbb{R}$, interpretable across models and texts.

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Metric Evaluation of FG score

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The objective is assess the explanatory power of semantic Forward Guidance (FG) indicators with respect to macro-financial variables (e.g., systemic stress). For this, we use the approach by Ridge Regression.

- Predict a target variable y_t (e.g., financial stress) from one or more FG scores $s_t^{(i)}$.
- Estimate $\hat{y}_t = \sum_i \beta_i s_t^{(i)}$ with ℓ_2 regularization:

$$\hat{\beta} = \arg \min_{\beta} \sum_t (y_t - \mathbf{s}_t^{\top} \beta)^2 + \lambda \|\beta\|_2^2$$

- Compare FG indicators from different models (e.g., ModernBERT, Mamba) using R^2 proportion of explained variance.

Schema Auto-Encoder

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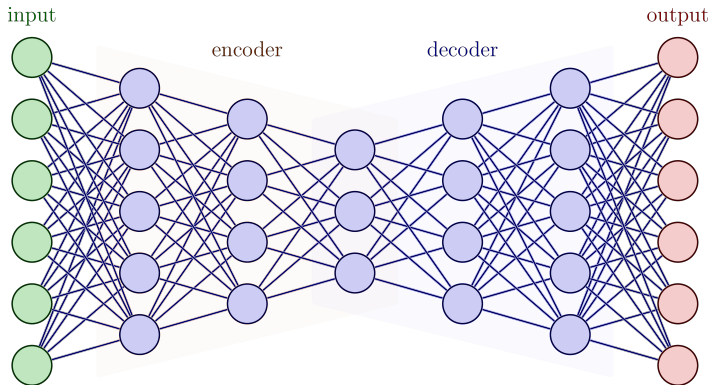


Figure: Schema Auto-Encoder.

Encoding and latent space

We have multivariate time series $\{x_t \in \mathbb{R}^d\}_{t=1}^T$ and, optionally, an exogenous variable c_t . We then construct sliding windows of length L :

$$X_i = [x_i, x_{i+1}, \dots, x_{i+L-1}] \in \mathbb{R}^{L \times d}, \quad C_i = c_{i+L-1} \in \mathbb{R}, \text{ for } i = 1, \dots, N \text{ with } N = T - L + 1.$$

Each window X_i is encoded into a hidden representation by an LSTM encoder:

$$h_i = \text{LSTM}_\phi(X_i) \in \mathbb{R}^h.$$

If we have an exogenous control signal C_i , we concatenate it to the LSTM's output:

$$v_i = \begin{cases} [h_i; C_i] \in \mathbb{R}^{h+1}, & \text{(conditional)} \\ h_i \in \mathbb{R}^h, & \text{(non-conditional)} \end{cases}$$

Finally, this vector is projected into the k -dimensional latent space by a linear layer (the encoder's final affine mapping):

$$z_i = W_\phi v_i + b_\phi \in \mathbb{R}^k.$$

We enforce the empirical distribution $\{z_i\}$ to align with the prior $p(z) = \mathcal{N}(0, I)$ in the sense of entropically-regularized Wasserstein-2. Given a batch of size B (Tolstikhin et al., 2019).

Record on Wasserstein-2 and Sinkhorn

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For two distributions p, q on $\mathbb{R}^k$, the Wasserstein-2 distance is defined by the Kantorovich transport problem:

$$W_2^2(p, q) = \min_{\Gamma \in \Pi(p, q)} \int_{\mathbb{R}^k \times \mathbb{R}^k} \|x - y\|^2 d\Gamma(x, y),$$

where Π is the :

$$\Pi(p, q) = \left\{ \Gamma \geq 0 \mid \forall A, \Gamma(A \times \mathbb{R}^k) = p(A), \Gamma(\mathbb{R}^k \times A) = q(A) \right\}$$

To make this problem computable and differentiable, we add an entropic regularization (Sinkhorn) term :

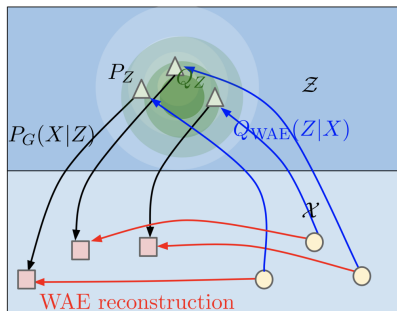
$$W_{2,\varepsilon}^2(p, q) = \min_{\Gamma \in \Pi(p, q)} \sum_{i,j} \Gamma_{ij} \|x_i - y_j\|^2 - \varepsilon H(\Gamma)$$

where $\Gamma = [\gamma_{ij}]$, $\forall \gamma_{ij} \geq 0$ and $\sum_{i,j} \gamma_{ij} = 1$ demonstration :

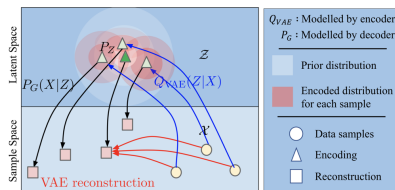
$$H(\Gamma) = - \sum_{i,j} \Gamma_{ij} \log \Gamma_{ij} = - \sum_{i,j} \gamma_{ij} \log \gamma_{ij}, \quad \varepsilon > 0 \text{ (régularization)}.$$

Wasserstein distance vs Kullback-Leiber

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(a) Wasserstein Auto-Encoder reconstruction



(b) Variational Auto-Encoder reconstruction

Discrete Sinkhorn-Wasserstein on a Batch

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Given a batch of encoded latents $\{z_i\}_{i=1}^B \sim q_\phi(z)$ and B samples $\{z'_j\}_{j=1}^B \sim p(z) = \mathcal{N}(0, I)$. We can define the cost matrix :

$$C_{ij} = \|z_i - z'_j\|^2 \quad \text{for } i, j = 1, \dots, B$$

Enforce uniform marginals

$$\Gamma \mathbf{1}_B = \frac{1}{B} \mathbf{1}_B, \quad \Gamma^T \mathbf{1}_B = \frac{1}{B} \mathbf{1}_B$$

Entropically-regularized OT problem:

$$W_{\varepsilon,2}^2 = \min_{\Gamma \in \mathbb{R}_+^{B \times B}} \sum_{i,j} \Gamma_{ij} C_{ij} - \varepsilon \sum_{i,j} \Gamma_{ij} \log \Gamma_{ij}$$

Sinkhorn–Knopp Algorithm

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Algorithm 0: Sinkhorn–Knopp Iterative Scaling Algorithm

Input: Cost matrix $C \in \mathbb{R}^{B \times B}$, regularization $\varepsilon > 0$, iterations T

Output: Regularized transport plan Γ and Sinkhorn cost $W_{\varepsilon,2}^2$

- 1 Compute the Gibbs kernel **demonstration**:

$$K_{ij} = \exp(-C_{ij}/\varepsilon)$$

- 2 Initialize scaling vectors:

$$u^{(0)} = v^{(0)} = \mathbf{1}_B$$

- 3 **for** $t = 0$ **to** $T - 1$ **do**

4

$$u^{(t+1)} = a \oslash (K v^{(t)}), \quad v^{(t+1)} = b \oslash (K^T u^{(t+1)})$$

- 5 Form transport plan:

$$\Gamma = \text{diag}(u^{(T)}) K \text{diag}(v^{(T)})$$

- 6 Compute Sinkhorn cost:

$$W_{\varepsilon,2}^2 = \sum_{i,j} \Gamma_{ij} C_{ij}$$

Regularization Term in WAE–Sinkhorn Loss

The latent vector z_i (and C_i if conditional) is then repeated L times and fed into the decoder:

$$\tilde{v}_i = \begin{cases} \begin{bmatrix} z_i; C_i \end{bmatrix} \in \mathbb{R}^{k+1}, & \text{(conditional)} \\ z_i \in \mathbb{R}^k, & \text{(non-conditional)} \end{cases} \quad U_i = \text{Repeat}(\tilde{v}_i, L) \in \mathbb{R}^{L \times \dim},$$

$$\hat{X}_i = \text{LSTM}_\theta(U_i), \quad \hat{X}_i(t) = W'_\theta[\text{LSTM_hidden}_t] + b'_\theta.$$

The total function loss is :

$$\mathcal{L} = \underbrace{\frac{1}{B} \sum_{i=1}^B \|X_i - \hat{X}_i\|^2}_{\text{Reconstruction MSE}} + \underbrace{\lambda W_{\varepsilon,2}^2(q_\phi(z), p(z))}_{\text{Sinkhorn OT penalty}}$$

- $W_{\varepsilon,2}^2(q, p) = \min_{\Gamma \in \Pi_B} \sum_{i,j} \Gamma_{ij} \|z_i - z'_j\|^2 - \varepsilon H(\Gamma)$
- $\Pi_B = \{\Gamma \geq 0 : \Gamma \mathbf{1} = \frac{1}{B} \mathbf{1}, \Gamma^T \mathbf{1} = \frac{1}{B} \mathbf{1}\}$, in practice: $\lambda = 1$.

Remarks:

- $\varepsilon \rightarrow 0 \Rightarrow$ approaches true W_2^2 , but may cause numerical instability.
- ε large $\Rightarrow$ smoother transport plan, faster convergence, but less geometric fidelity.

New vision for errors in econometrics

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Stress as a latent dimension.

If one adopts the reconstruction error e_i as a proxy for the level of financial stress at time i , then:

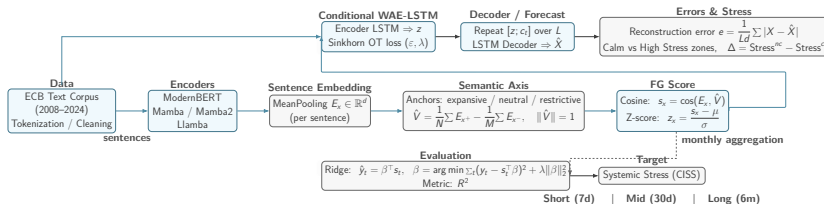
$$\text{Stress}_i \equiv e_i \left(\text{où } e_i = \frac{1}{Ld} \sum_{t=1}^L \sum_{j=1}^d |X_i(t, j) - \hat{X}_i(t, j)| \right)$$

Two zones can be defined: calm zone $\text{Stress}_i \leq T_\alpha$ and High stress zone: $\text{Stress}_i > T_\alpha$.

- $\text{Stress}_i^{\text{nc}}$ (blue): “idiosyncratic” stress without Forward Guidance (FG), the hypothetical stress level had no FG signal been issued.
- $\text{Stress}_i^{\text{c}}$ (orange): “residual” stress after accounting for FG
- $\Delta_i = \text{Stress}_i^{\text{nc}} - \text{Stress}_i^{\text{c}}$, the quantity Δ_i can be interpreted as the marginal contribution of Forward Guidance (FG) to systemic stress reduction at time i . It captures the portion of financial stress that can be attributed to the absence of FG, by comparing the latent stress level without FG to the residual stress after FG has been accounted for.

Overview of the methodology

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Flow: text → embeddings → semantic axis → FG score

Models: ModernBERT, Mamba/Mamba2, Llama (embeddings)

Evaluation: Ridge vs CISS | **Generative:** Conditional WAE-LSTM + Sinkhorn OT

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Model Comparison: FG Scores via Ridge Regression

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Embedding Model	Input Type	FG Score Dim.	R^2 (Ridge)
Mamba (130M)	Sentence mean pooling	$\mathbb{R}^{768}$	0.77
Llama (7B)	Sentence mean pooling	$\mathbb{R}^{4096}$	0.68
ModernBERT	Sentence mean pooling	$\mathbb{R}^{768}$	0.58

Best performance with Mamba embeddings for predicting systemic stress.

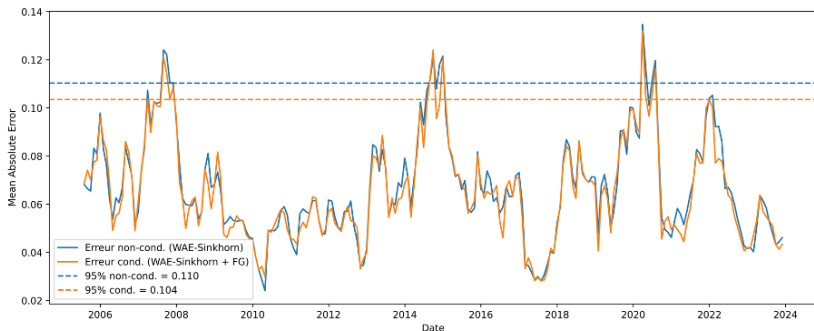


Figure: Superimposed reconstruction errors of the conditional and non-conditional models (short term).

In the short term (7-day horizon), latent stress is generally well anticipated, and Forward Guidance replicates systemic stress without notable attenuation.

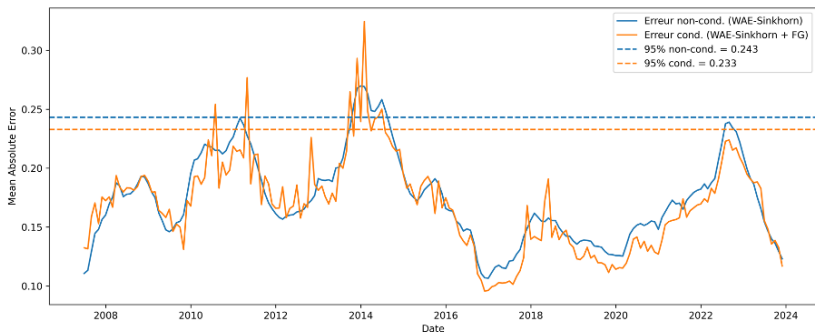


Figure: Superimposed reconstruction errors of the conditional and non-conditional models (mid term).

At the medium-term horizon (30 days), the conditional model with Forward Guidance shows that during crisis periods, communication becomes erratic, with higher latent stress.

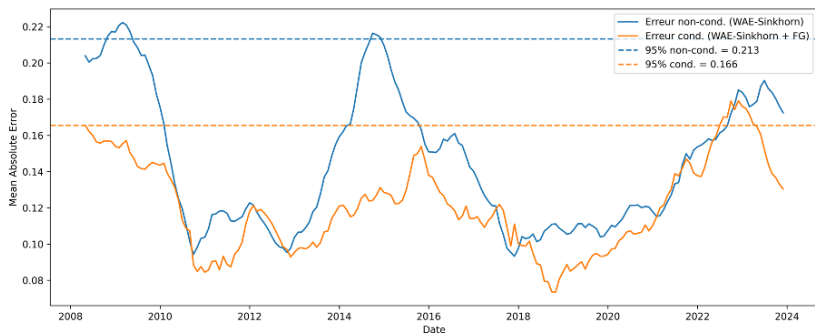


Figure: Superimposed reconstruction errors of the conditional and non-conditional models (long term).

In the long term (6 months), a communication lag of several months can be observed between the ECB and systemic stress.

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■ Alternative OT divergences:

- Sinkhorn divergences (subtract self-transport terms) for unbiased estimates.
- Unbalanced OT to handle batch size mismatches or missing data.

■ Richer decoders / temporal models:

- Replace LSTM with Transformer or Temporal Convolution for faster parallel training.
- Add attention over latent+control for improved sequence reconstruction.

■ Exogenous signals & multimodality:

- Incorporate additional macro indicators (CPI, unemployment, market liquidity).
- Multimodal extension: combine text-based FG embeddings with numeric data.

**THANK YOU FOR YOUR
ATTENTION !**

Questions & Discussion

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References II

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Selective State Space Model Schema

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Selective State Space Model with Hardware-aware State Expansion

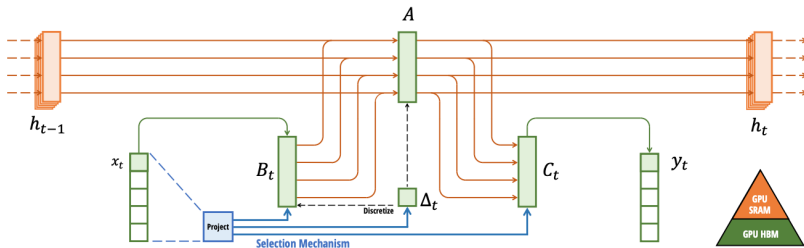


Figure: Selective State Space Model Mechanism.

Proof : Relative Entropy [back](#)

Let $\gamma \in \Pi(p, q)$ and $\mu = p \otimes q$, with $\gamma \ll \mu$. By Radon–Nikodym there exists

$$f(x, y) = \frac{d\gamma}{d\mu}(x, y), \quad \int h d\gamma = \int h f d\mu.$$

The Kullback–Leibler divergence is

$$\text{KL}(\gamma \parallel \mu) = \int \log(f(x, y)) d\gamma(x, y),$$

and the *relative entropy* (negative KL) becomes

$$H(\gamma \parallel \mu) := -\text{KL}(\gamma \parallel \mu) = - \int \log(f(x, y)) d\gamma(x, y)$$

In particular, with $\mu = p \otimes q$ on $\mathbb{R}^d \times \mathbb{R}^d$, $H(\gamma \parallel \mu)$ measures how far γ deviates from independent marginals.

$$H(\gamma) = - \int \log \left(\frac{d\gamma}{d(p \otimes q)}(x, y) \right) d\gamma(x, y)$$

Proof : Discretisation of Relative Entropy

Let:

- $p = (a_i)_{i=1}^n$ be a discrete probability distribution over the points $\{x_i\} \subset \mathbb{R}^d$,
- $q = (b_j)_{j=1}^m$ be a discrete probability distribution over the points $\{y_j\} \subset \mathbb{R}^d$,
- and $\gamma = (\gamma_{ij}) \in \mathbb{R}^{n \times m}$, a discrete transport plan between p and q .

Then:

$$p \otimes q = \sum_{i=1}^n \sum_{j=1}^m a_i b_j \delta_{(x_i, y_j)} \quad \text{and} \quad \gamma = \sum_{i=1}^n \sum_{j=1}^m \gamma_{ij} \delta_{(x_i, y_j)}$$

At each atomic point (x_i, y_j) , the Radon–Nikodym derivative (i.e., the density) of γ with respect to $p \otimes q$ is given by:

$$\frac{d\gamma}{d(p \otimes q)}(x_i, y_j) = \frac{\gamma_{ij}}{a_i b_j} \quad (\text{since } (p \otimes q)(x_i, y_j) = a_i b_j)$$

The continuous formula thus becomes:

$$H(\gamma) = - \sum_{i,j} \gamma_{ij} \log \left(\frac{\gamma_{ij}}{a_i b_j} \right)$$

Proof : Discretisation of Relative Entropy

We expand:

$$H(\gamma) = - \sum_{i,j} \gamma_{ij} \log \gamma_{ij} + \sum_{i,j} \gamma_{ij} \log(a_i b_j) = - \sum_{i,j} \gamma_{ij} \log \gamma_{ij} + \sum_i \log a_i \sum_j \gamma_{ij} + \sum_j \log b_j \sum_i \gamma_{ij}$$

Using the marginal constraints: $\sum_j \gamma_{ij} = a_i$, $\sum_i \gamma_{ij} = b_j$, we obtain:

$$\sum_i \log a_i \sum_j \gamma_{ij} = \sum_i \log a_i \cdot a_i = \sum_i a_i \log a_i, \quad \sum_j \log b_j \sum_i \gamma_{ij} = \sum_j \log b_j \cdot b_j = \sum_j b_j \log b_j$$

Therefore:

$$H(\gamma) = - \sum_{i,j} \gamma_{ij} \log \gamma_{ij} + \sum_i a_i \log a_i + \sum_j b_j \log b_j$$

Since the terms $\sum_i a_i \log a_i$ and $\sum_j b_j \log b_j$ are constant when optimizing over $\gamma \in \Pi(p, q)$, they can be ignored in the objective. We obtain the simplified discrete entropy form:

$$H(\Gamma) := - \sum_{i,j} \gamma_{ij} \log \gamma_{ij}$$

Proof : Wasserstein-Sinkhorn Algorithm [back](#)

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We consider two discrete probability distributions $a = (a_i)_{i=1}^n$ and $b = (b_j)_{j=1}^m$ on $\mathbb{R}^d$, supported on points $\{x_i\}, \{y_j\}$ respectively. The entropically regularized Wasserstein-2 distance is defined as:

$$W_{2,\varepsilon}^2(a, b) = \min_{\Gamma \in \Pi(a,b)} \sum_{i,j} \gamma_{ij} \|x_i - y_j\|^2 - \varepsilon \sum_{i,j} \gamma_{ij} \log \gamma_{ij} \quad (1)$$

where $\Pi(a, b)$ is the set of couplings $\Gamma \in \mathbb{R}_+^{n \times m}$ satisfying: $\sum_j \gamma_{ij} = a_i, \quad \sum_i \gamma_{ij} = b_j$

We introduce Lagrange multipliers α_i and β_j to enforce the marginal constraints. The Lagrangian becomes:

$$\mathcal{L}(\Gamma, \alpha, \beta) = \sum_{i,j} \gamma_{ij} \|x_i - y_j\|^2 - \varepsilon \sum_{i,j} \gamma_{ij} \log \gamma_{ij} + \sum_i \alpha_i \left(a_i - \sum_j \gamma_{ij} \right) + \sum_j \beta_j \left(b_j - \sum_i \gamma_{ij} \right)$$

To find the optimal Γ^* , we compute the stationary point of $\mathcal{L}$ with respect to each γ_{ij} :

$$\frac{\partial \mathcal{L}}{\partial \gamma_{ij}} = \|x_i - y_j\|^2 - \varepsilon(1 + \log \gamma_{ij}) - \alpha_i - \beta_j = 0$$

Proof : Wasserstein-Sinkhorn Algorithm

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Solving the equation:

$$\|x_i - y_j\|^2 - \varepsilon(1 + \log \gamma_{ij}) - \alpha_i - \beta_j = 0$$

leads to:

$$\log \gamma_{ij} = \frac{\alpha_i + \beta_j - \|x_i - y_j\|^2}{\varepsilon} - 1 \quad \Rightarrow \quad \gamma_{ij} = e^{-1} \cdot e^{\alpha_i/\varepsilon} \cdot e^{\beta_j/\varepsilon} \cdot e^{-\|x_i - y_j\|^2/\varepsilon}$$

Define:

$$u_i := e^{\alpha_i/\varepsilon}, \quad v_j := e^{\beta_j/\varepsilon}, \quad K_{ij} := \exp \left(-\frac{\|x_i - y_j\|^2}{\varepsilon} \right)$$

Then:

$$\boxed{\gamma_{ij} = u_i \cdot K_{ij} \cdot v_j} \quad \Rightarrow \quad \boxed{\Gamma = \text{diag}(u) K \text{diag}(v)}$$

Proof : Wasserstein-Sinkhorn Algorithm

We now impose the constraints $\Gamma \mathbf{1} = a$ and $\Gamma^\top \mathbf{1} = b$. Using the expression:

$$\Gamma = \text{diag}(u) K \text{diag}(v)$$

the constraints become:

$$\sum_j \gamma_{ij} = a_i \Rightarrow u_i (Kv)_i = a_i \Rightarrow u = \frac{a}{Kv}$$

$$\sum_i \gamma_{ij} = b_j \Rightarrow v_j (K^\top u)_j = b_j \Rightarrow v = \frac{b}{K^\top u}$$

This leads to the Sinkhorn-Knopp iterations :

$$\begin{cases} u^{(t+1)} = \frac{a}{Kv^{(t)}} \\ v^{(t+1)} = \frac{b}{K^\top u^{(t+1)}} \end{cases}$$

These iterations converge under mild conditions (e.g. $K_{ij} > 0$), providing the optimal Γ^* .